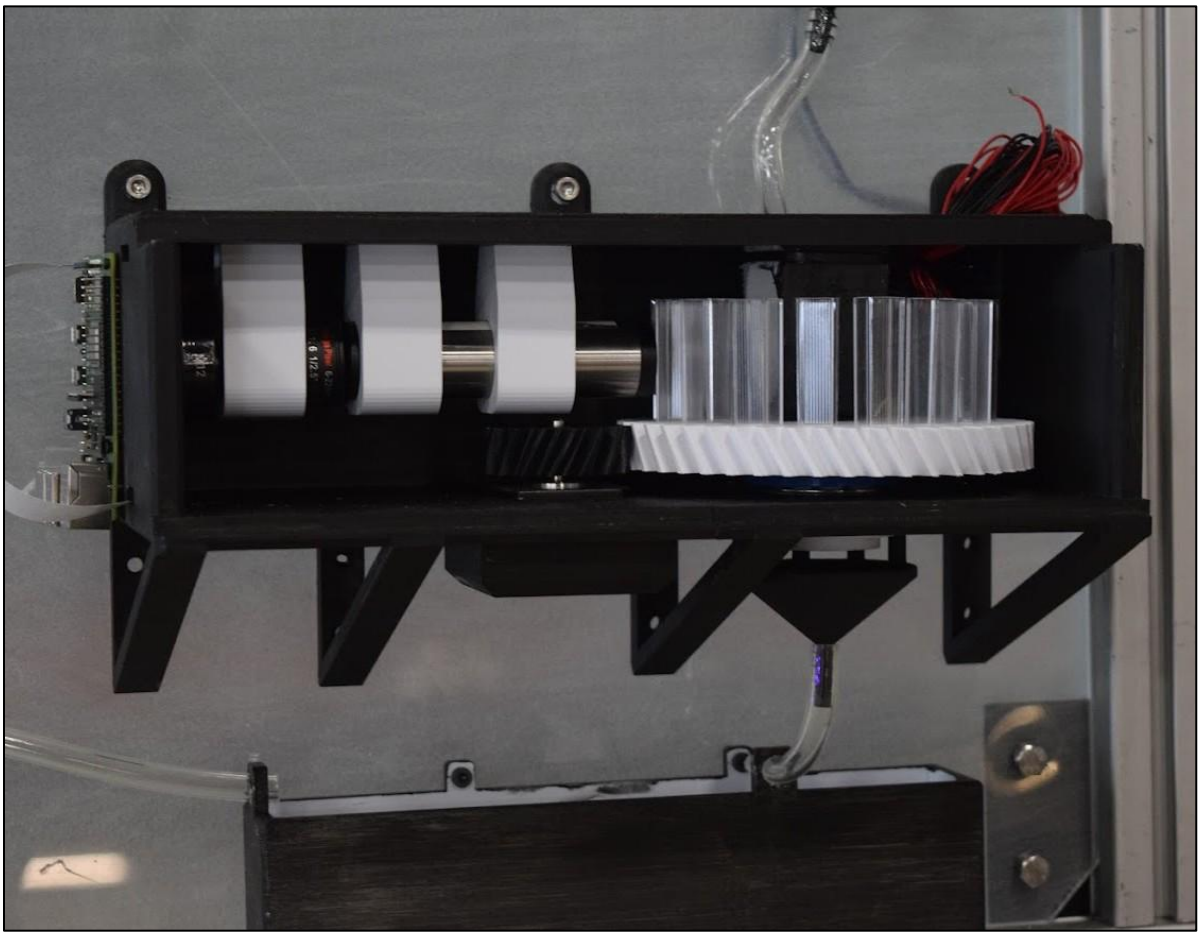


Figure 1: Final 3D-Printed and Fully Assembled Carousel

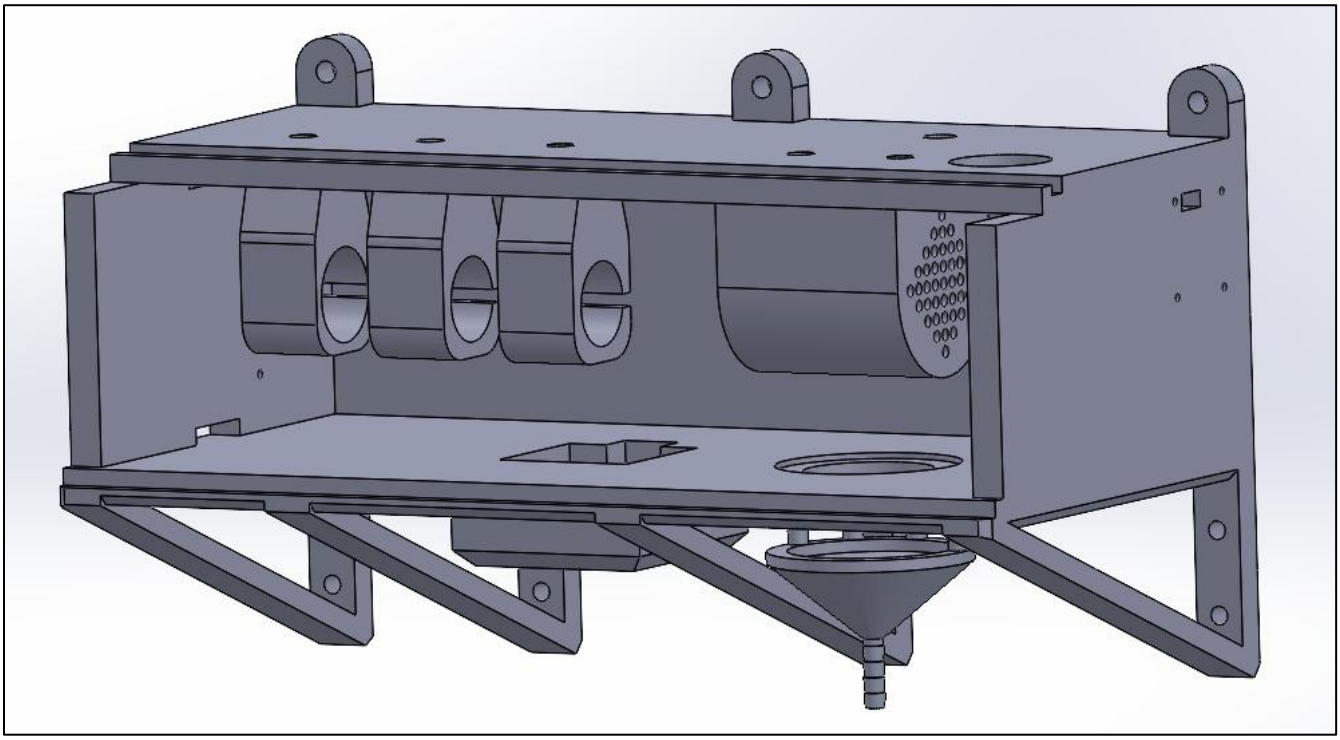


Figure 2: SolidWorks Model Without Additional Components

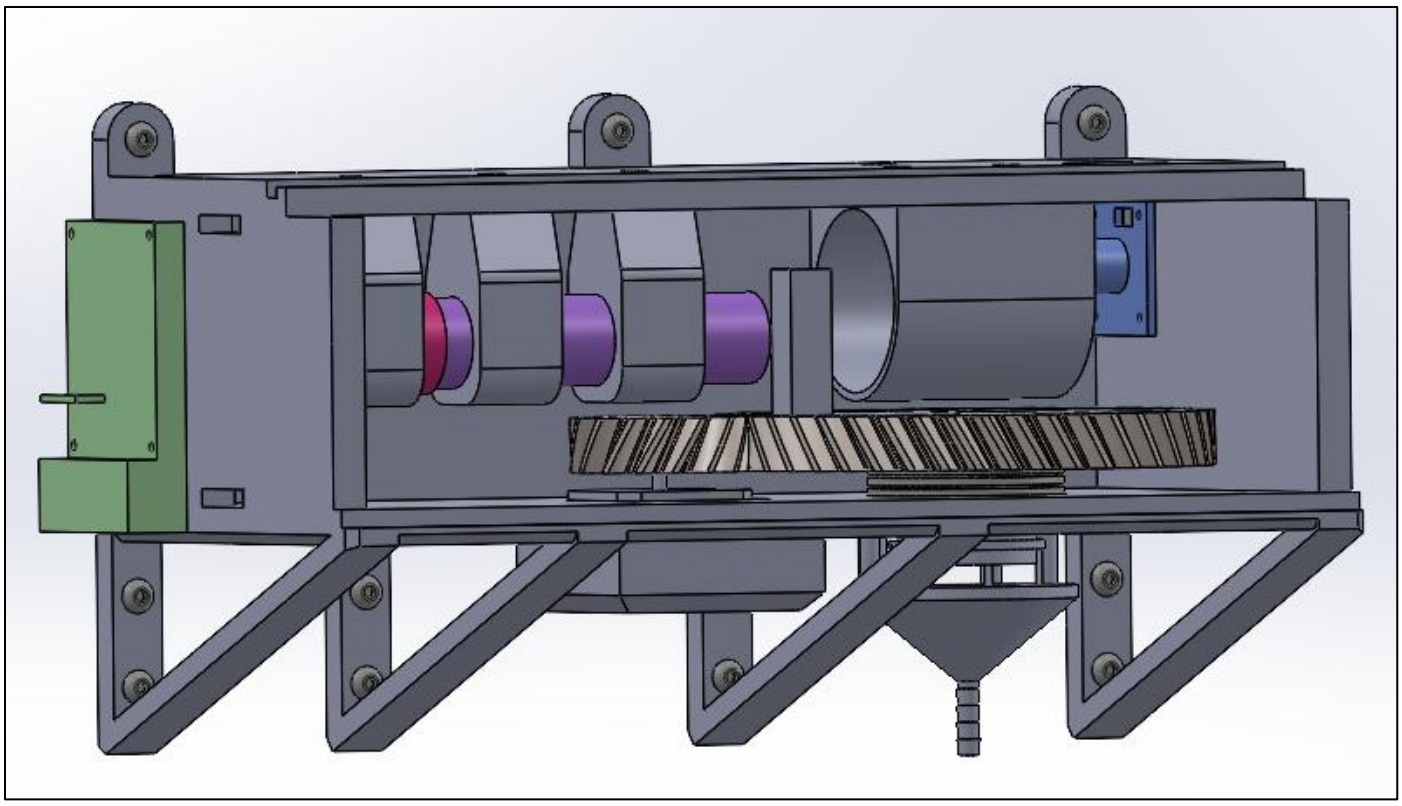


Figure 3: Left Perspective View of the SolidWorks Model with Integrated Components

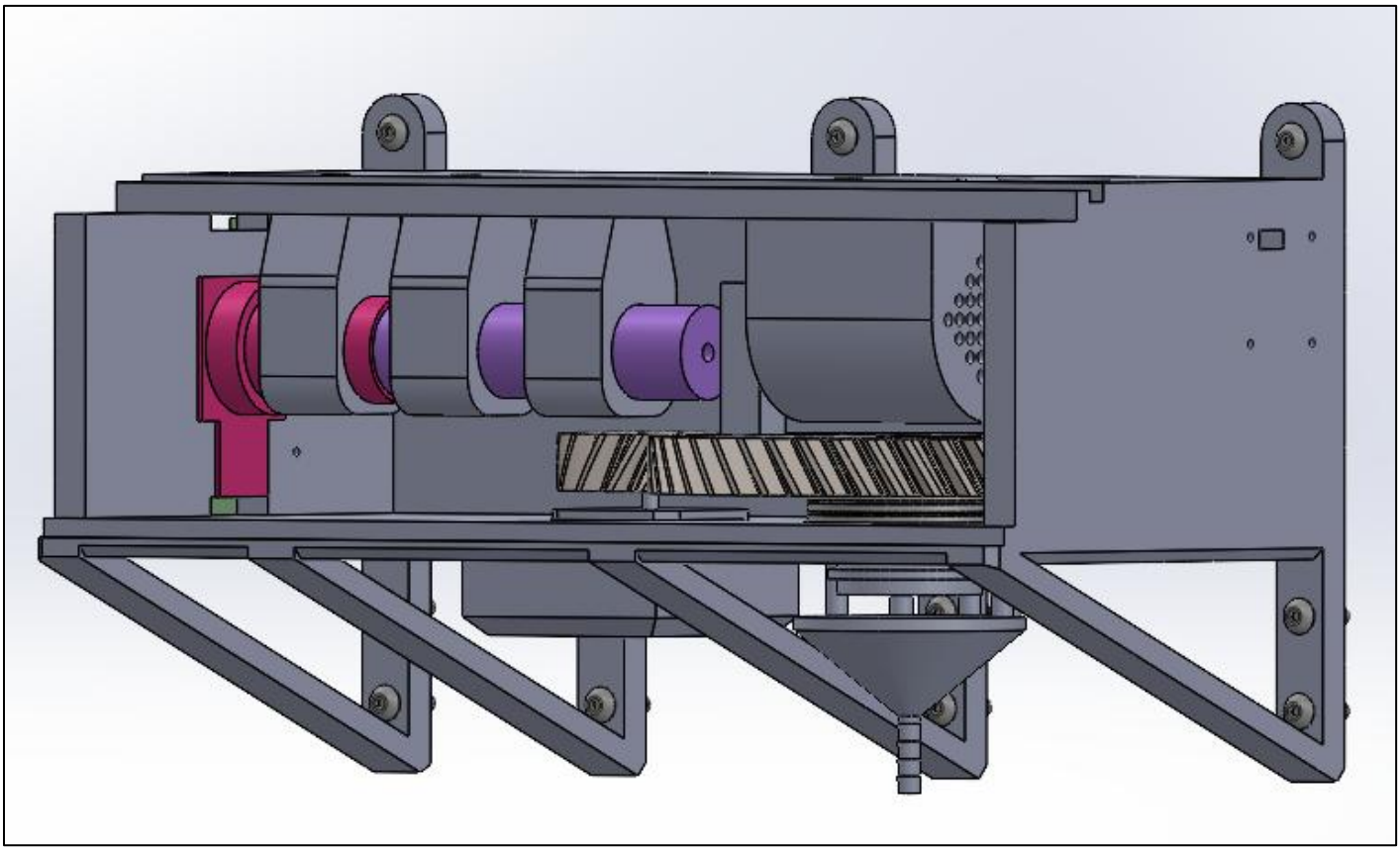


Figure 4: Right Perspective View of the SolidWorks Model with Integrated Components

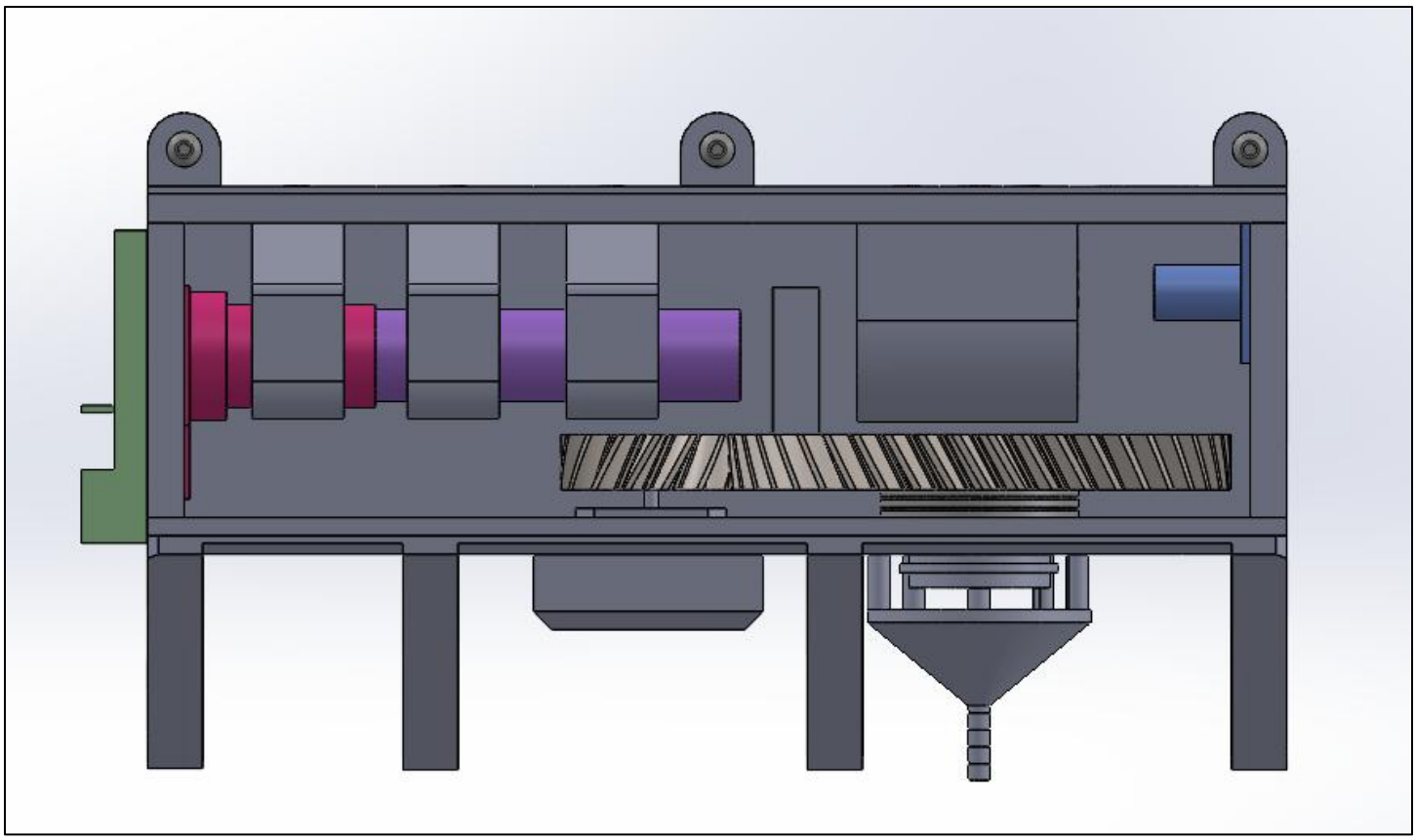


Figure 5: Front View of the SolidWorks Model with Integrated Components

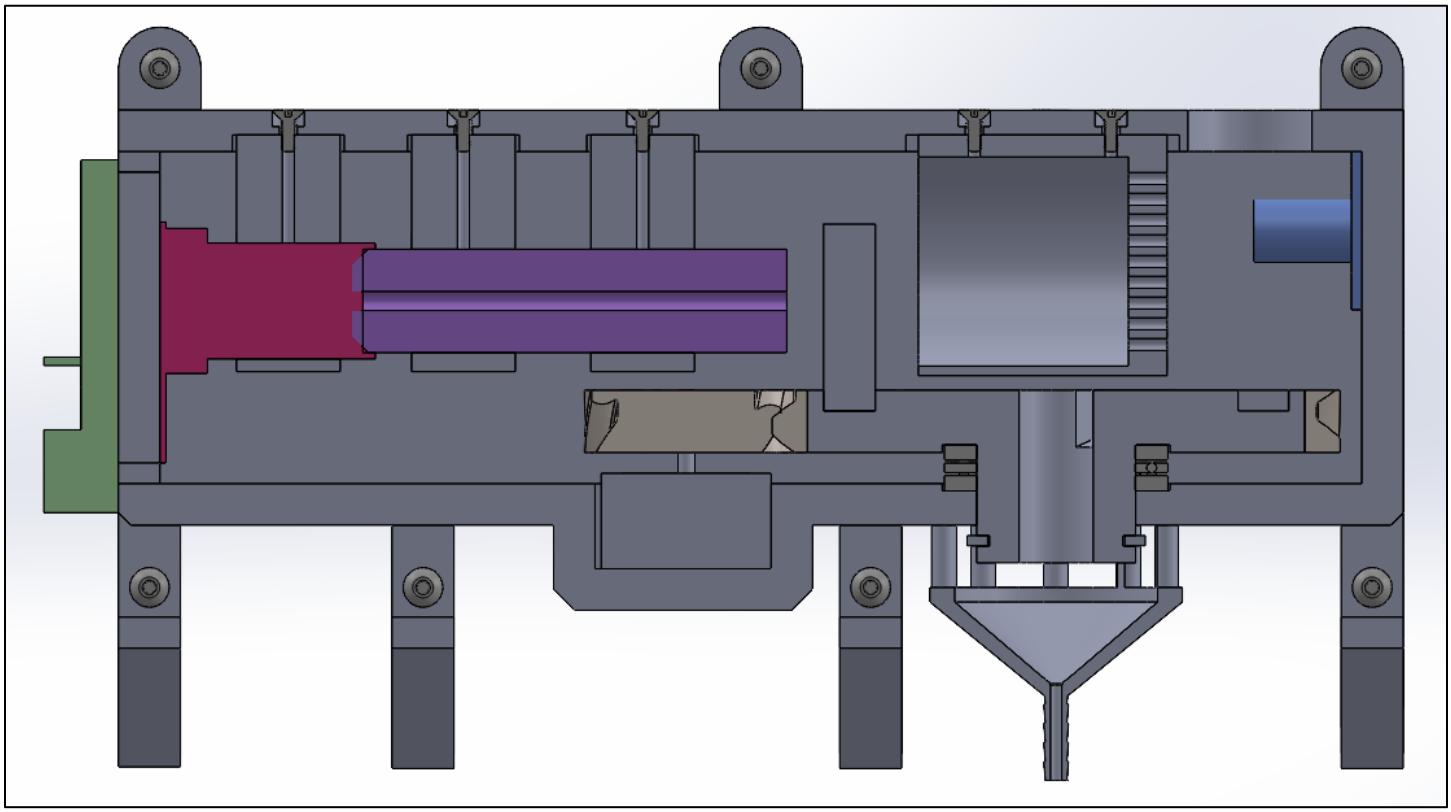


Figure 6: Section View of the SolidWorks Model with Integrated Components

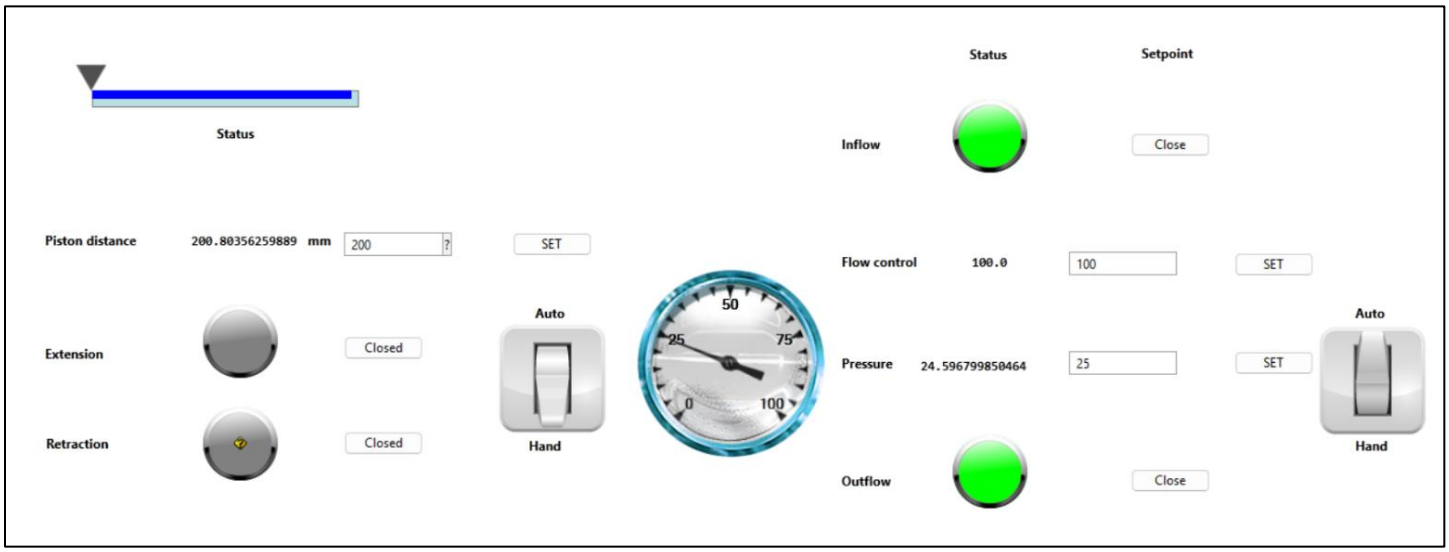


Figure 7: HMI Developed Using VTScada

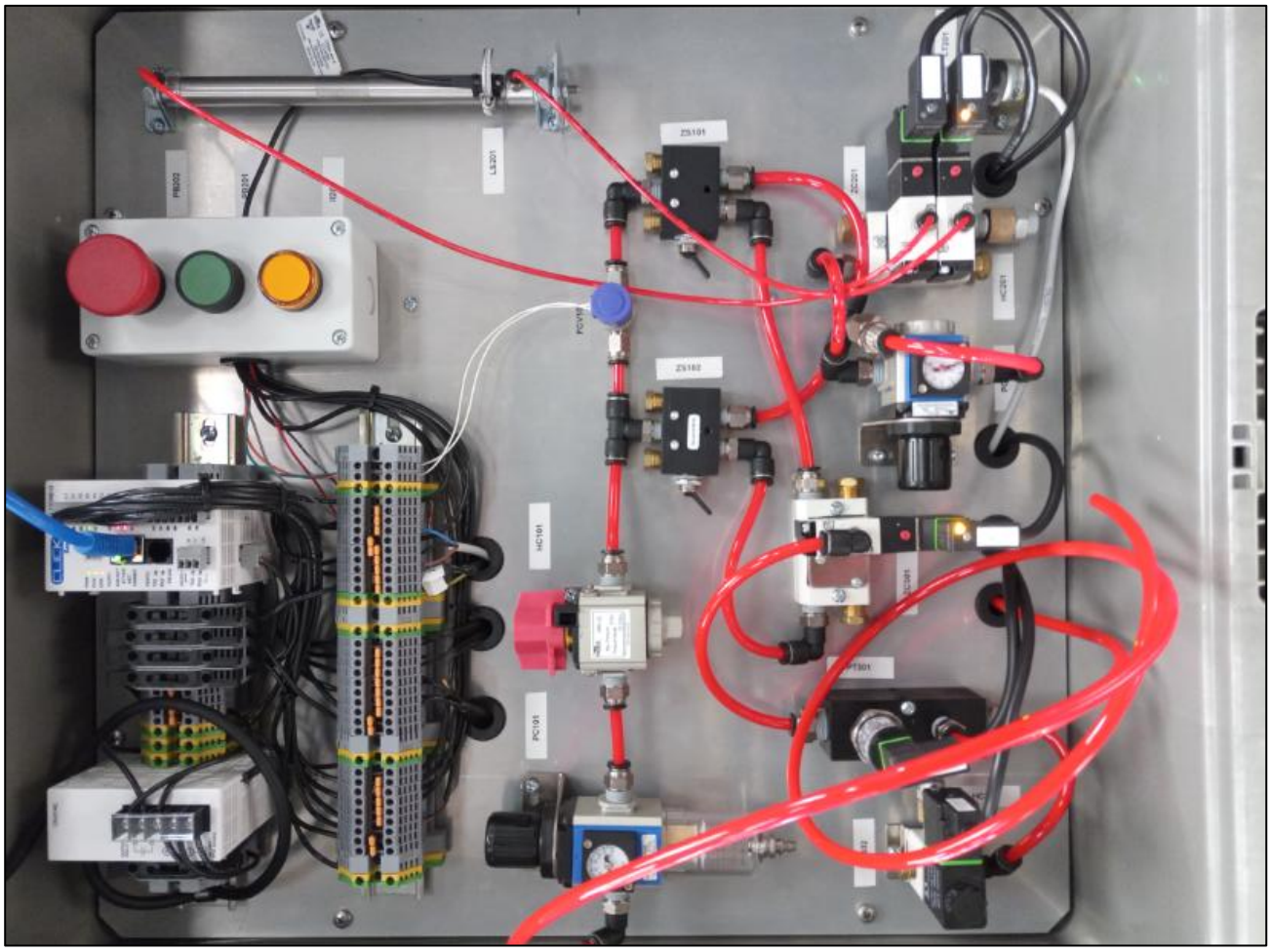


Figure 8: Industrial Automation Training Kit Used

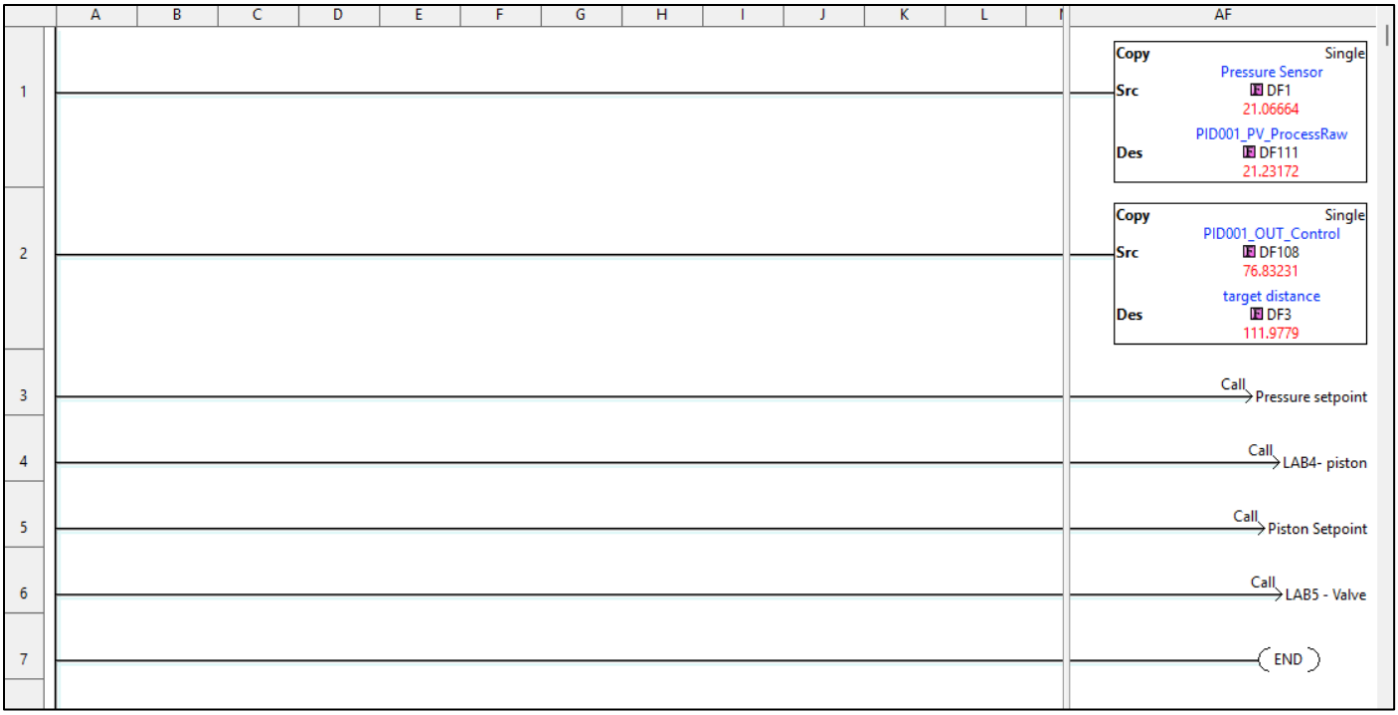


Figure 9: Main PLC Program Made with ClickPLC

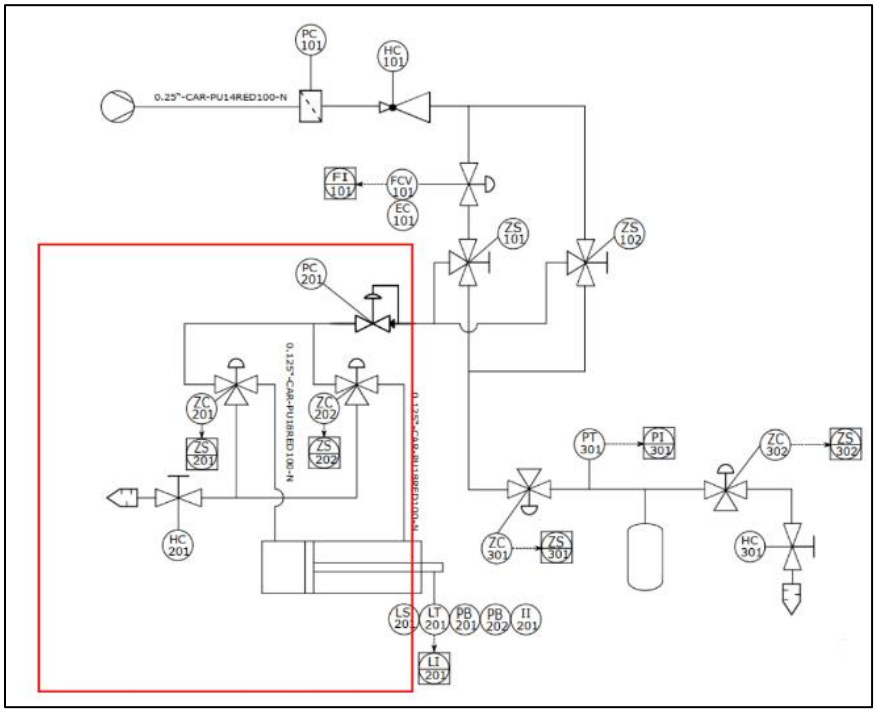


Figure 10: P&ID of the Lab Kit, Focused on Piston Control

- **Ultrasonic position transmitter (LT-201):** Measures piston position
- **Pneumatic valve (ZC-201):** Extends piston when actuated
- **Pneumatic valve (ZC-202):** Retracts position when actuated

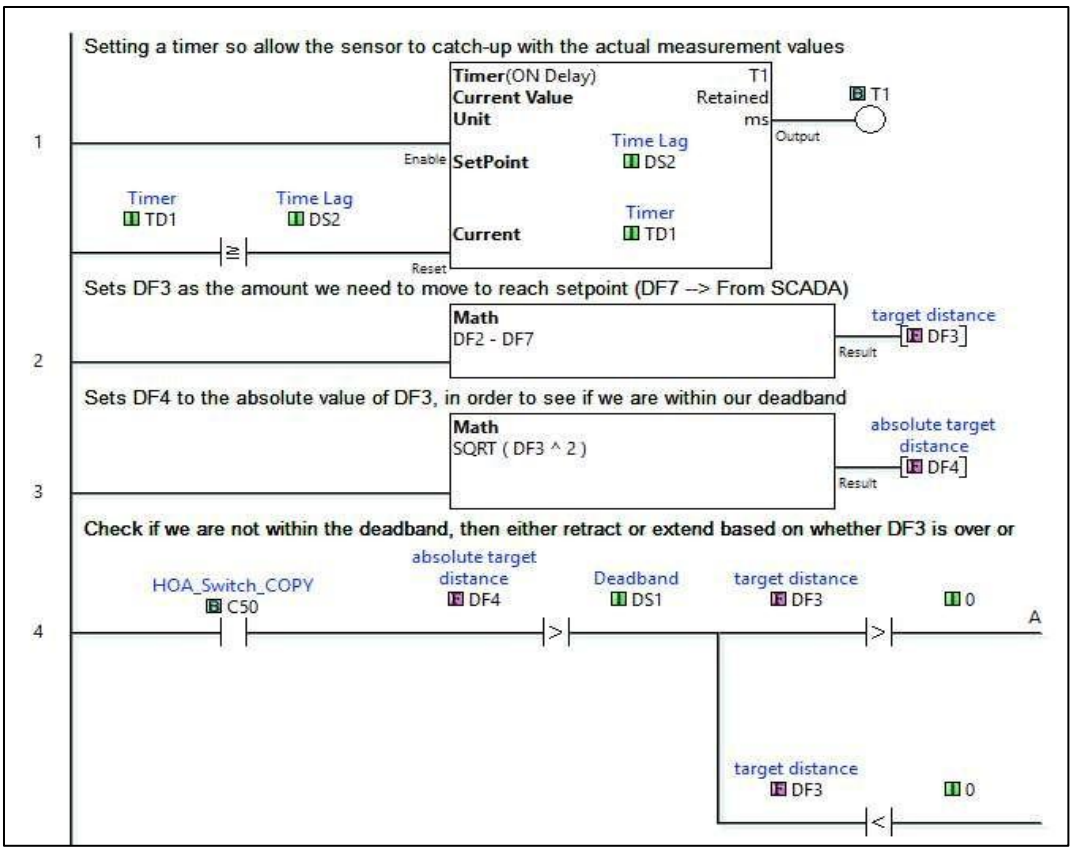


Figure 11: Piston Subroutine Program (1/2)

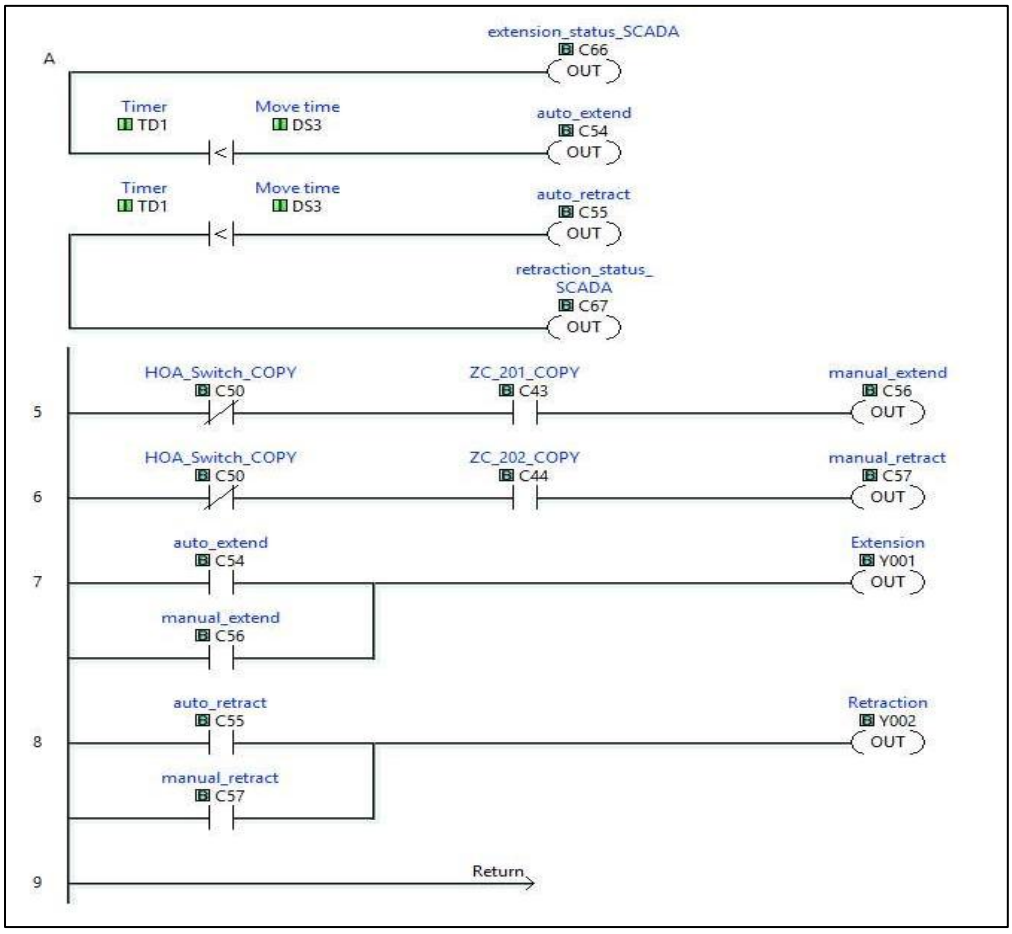


Figure 12: Piston Subroutine Program (2/2)

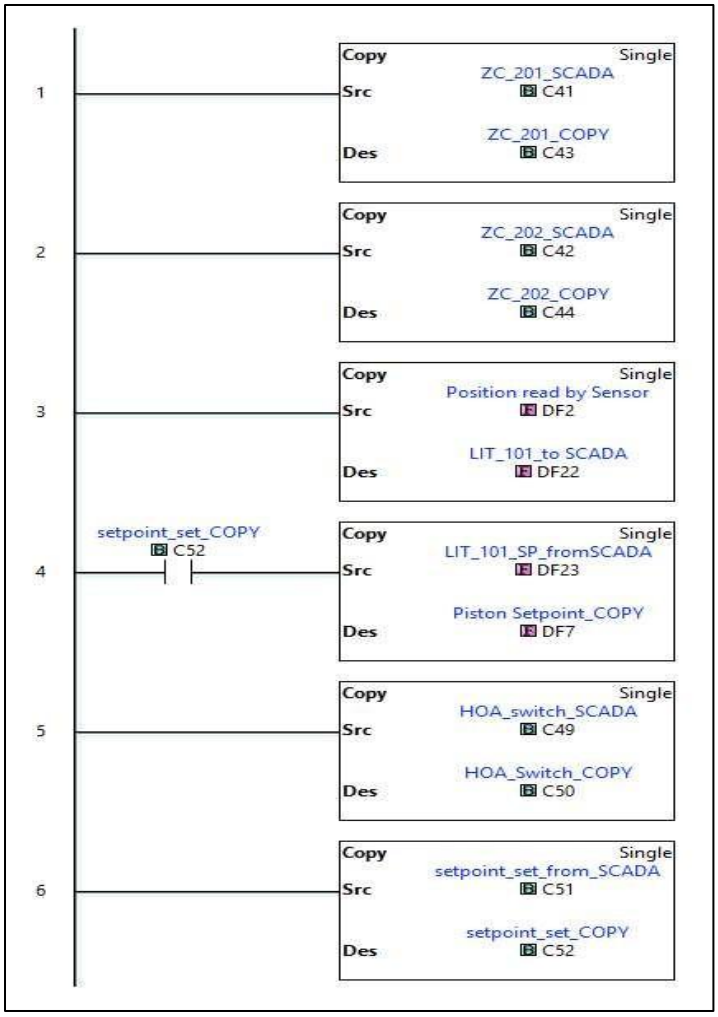


Figure 13: Piston Setpoint Subroutine Program (1/2)

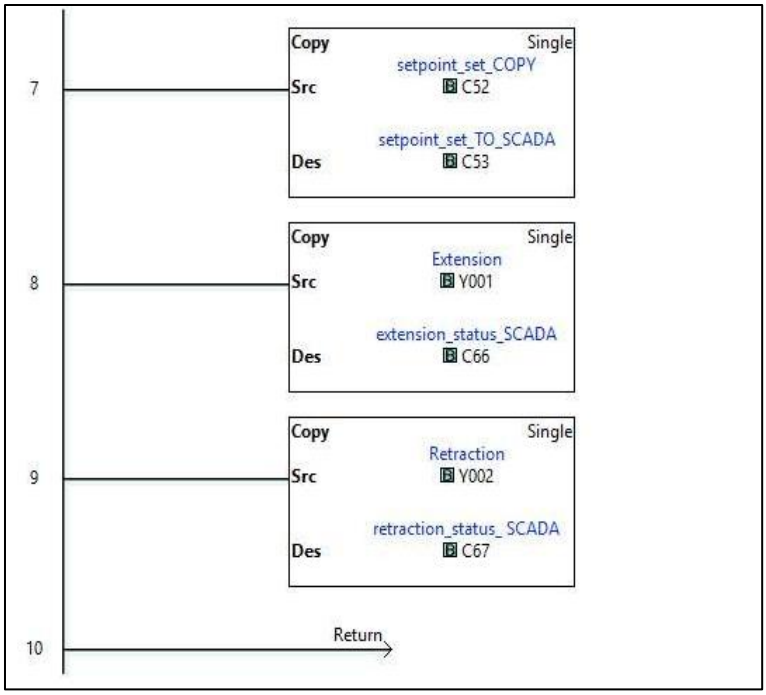


Figure 14: Piston Setpoint Subroutine Program (2/2)



Figure 15: Thermoformed Firetruck Shell

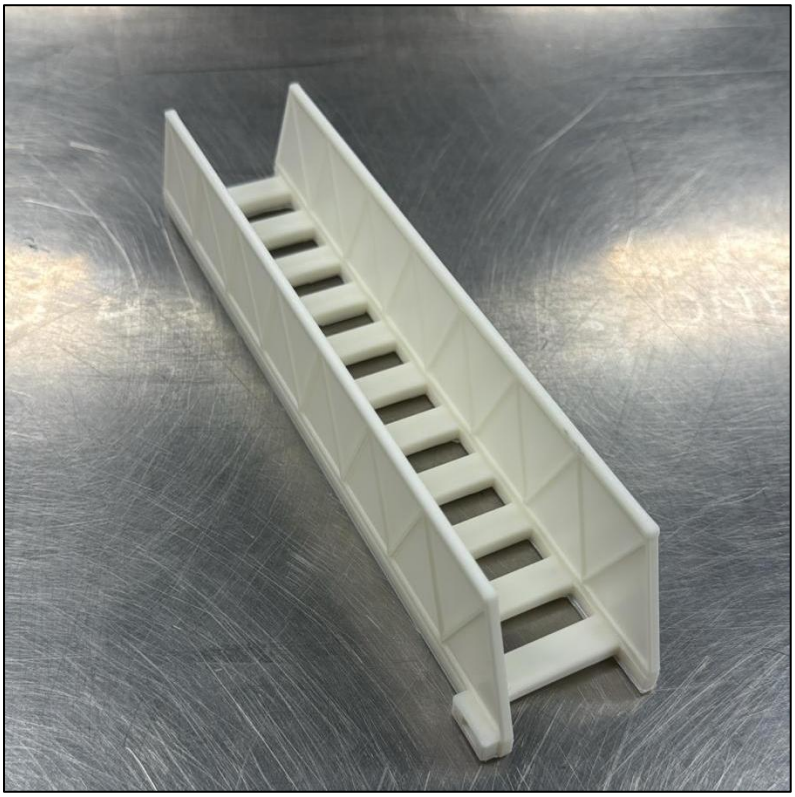


Figure 16: 3D-Printed Ladder

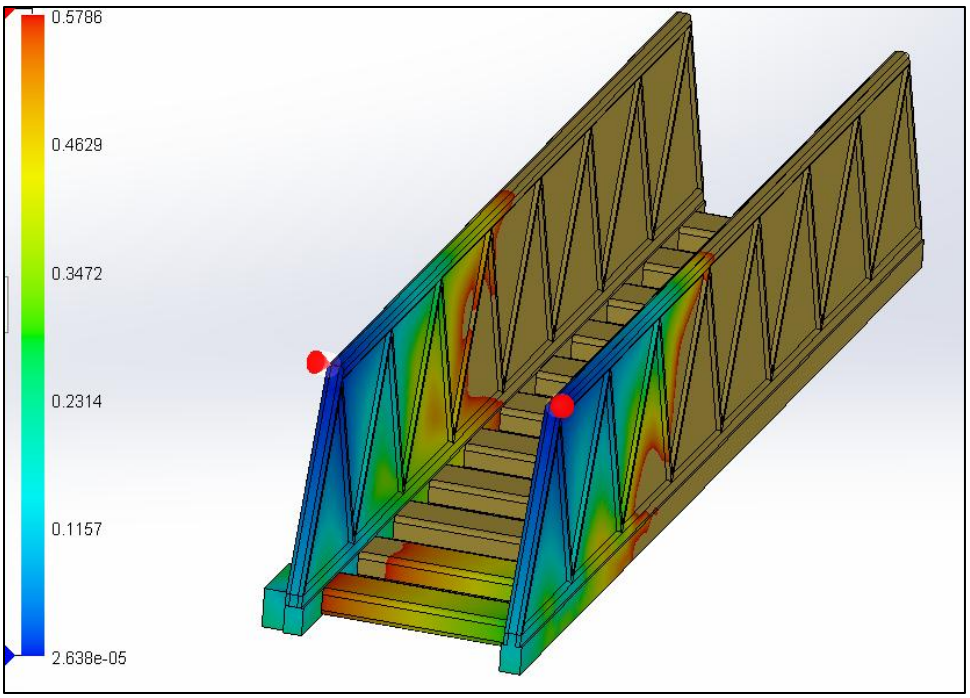


Figure 17: Injection Molding Simulation at 0.58 s

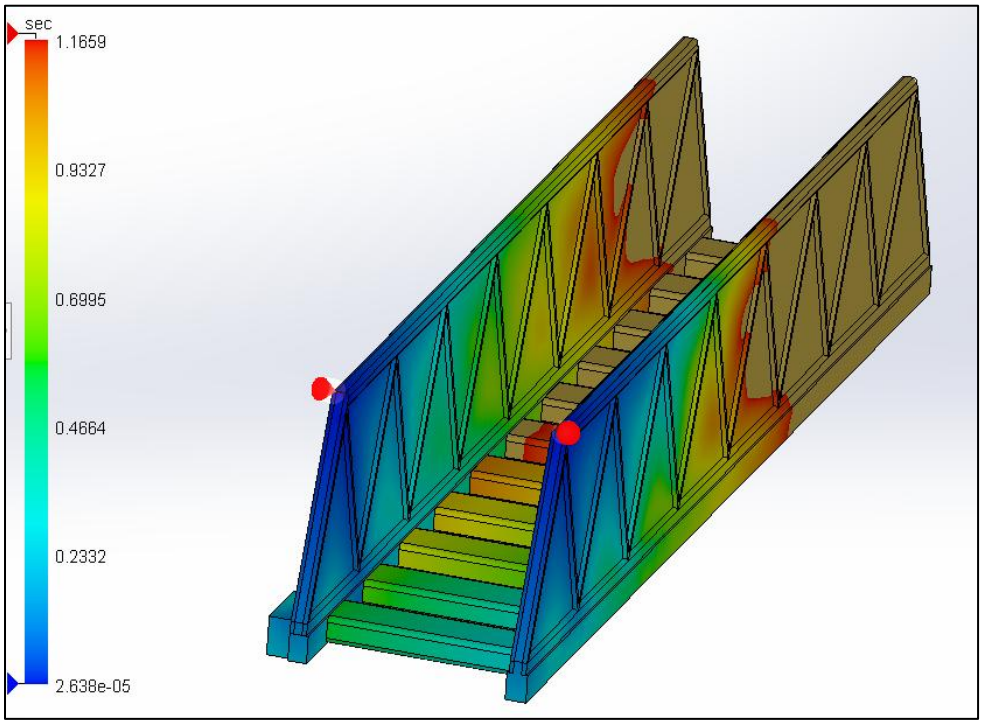


Figure 18: Injection Molding Simulation at 1.17 s

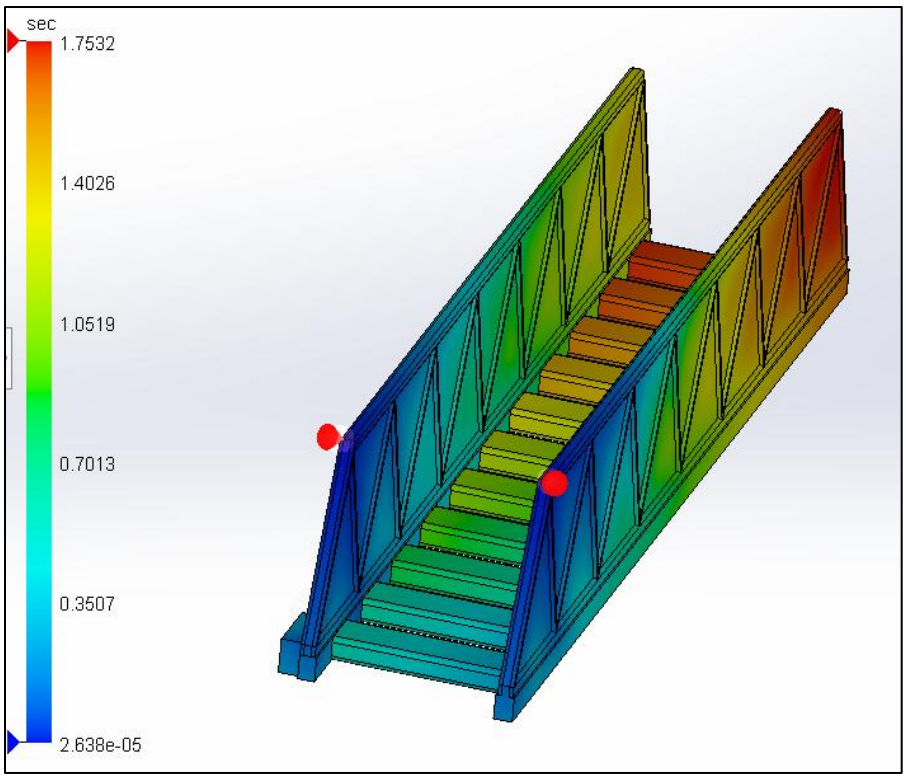


Figure 19: Injection Molding Simulation at 1.75 s

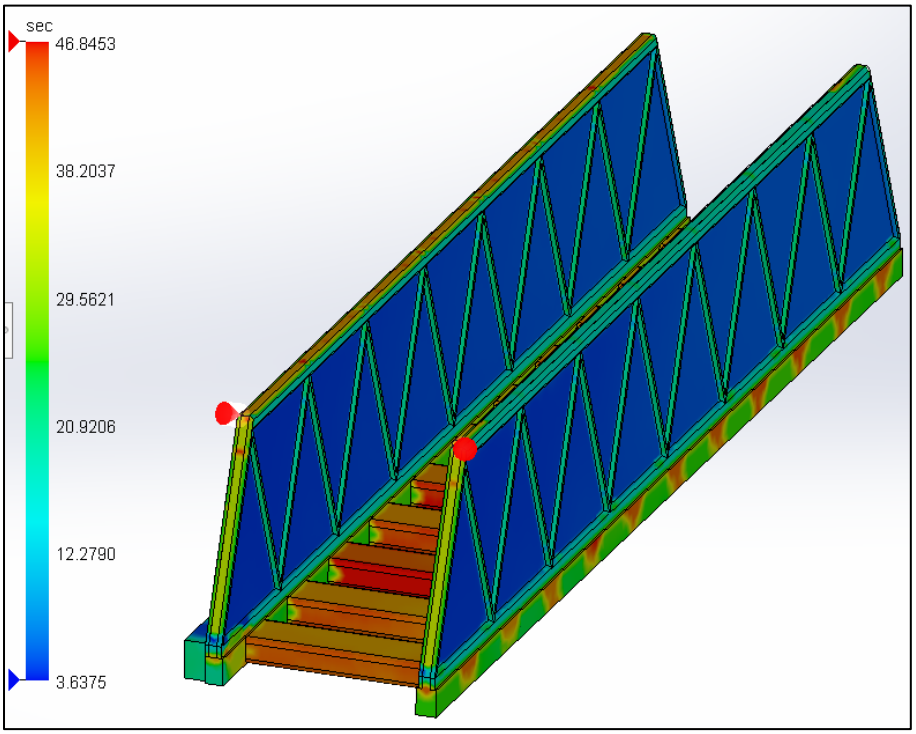


Figure 20: Cooling Time Analysis from SolidWorks Simulation

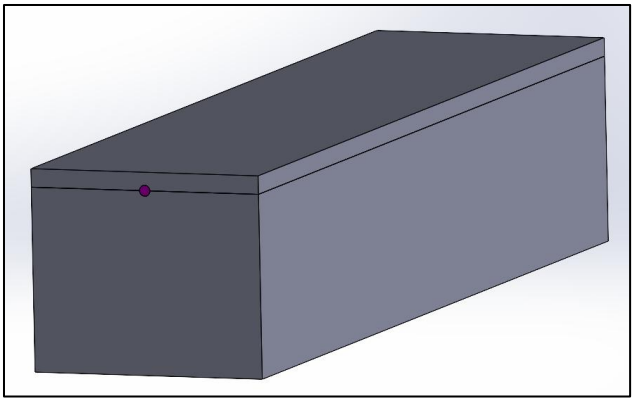


Figure 21: Closed Ladder Die

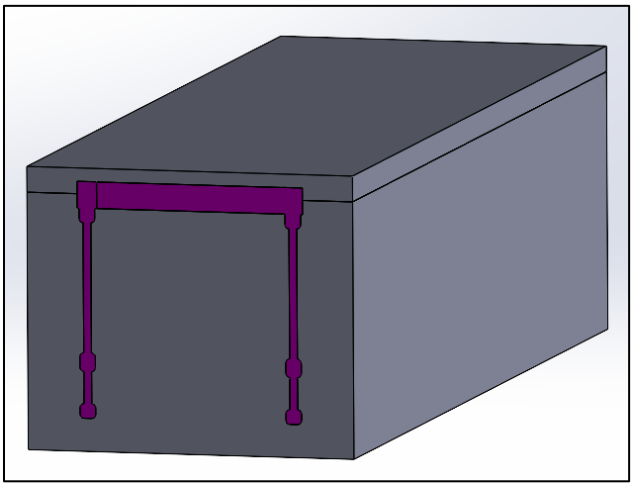


Figure 22: Section View of Ladder Die

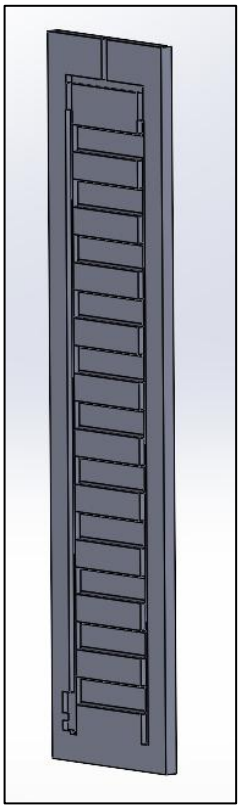


Figure 23: Top Half of Ladder Die

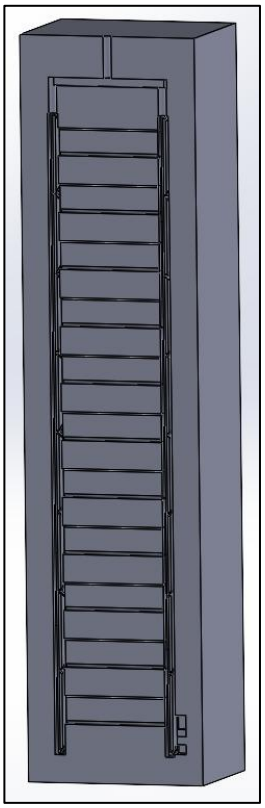


Figure 24: Bottom Half of Ladder Die

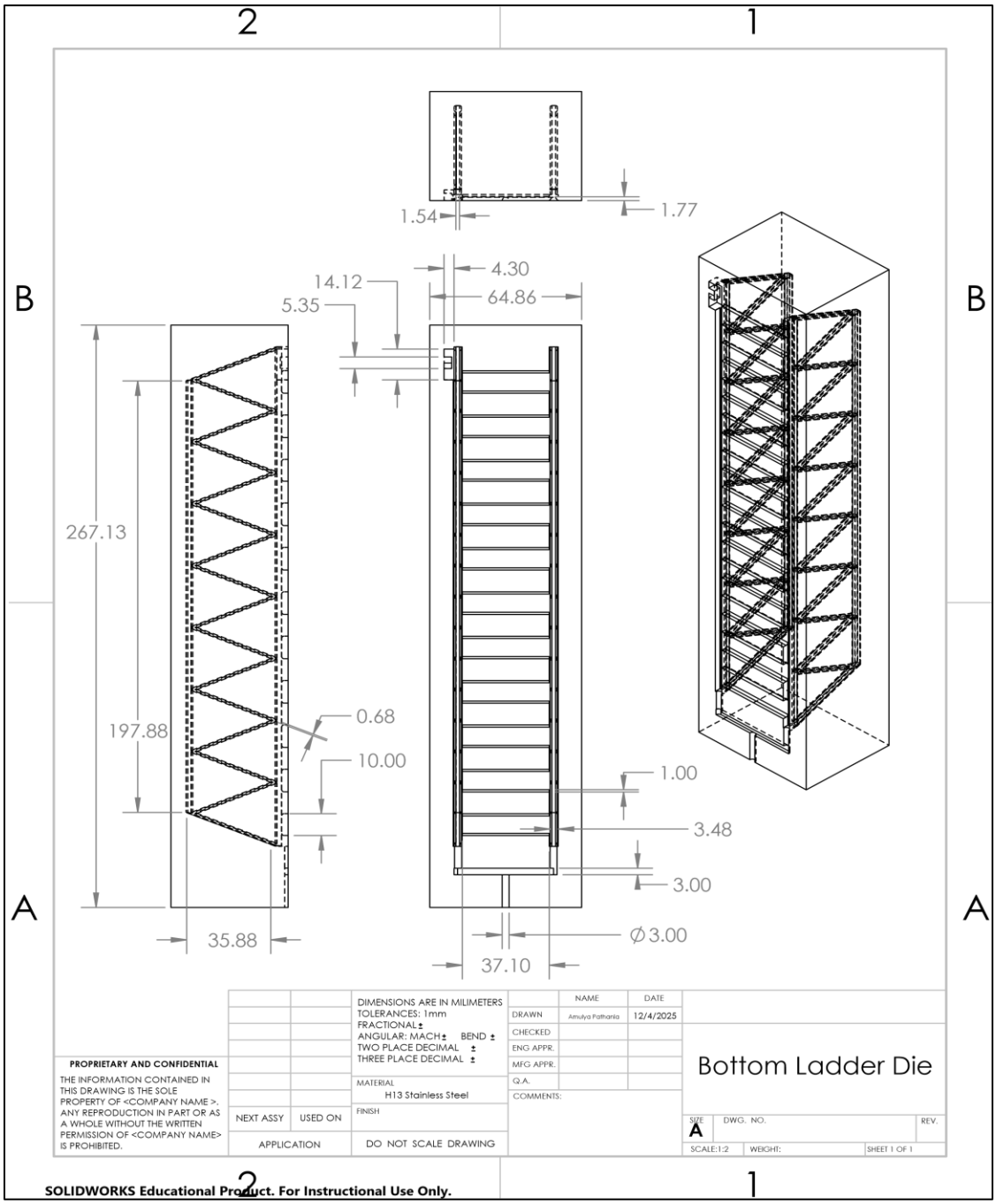


Figure 25: Engineering Drawing of Bottom Ladder Die

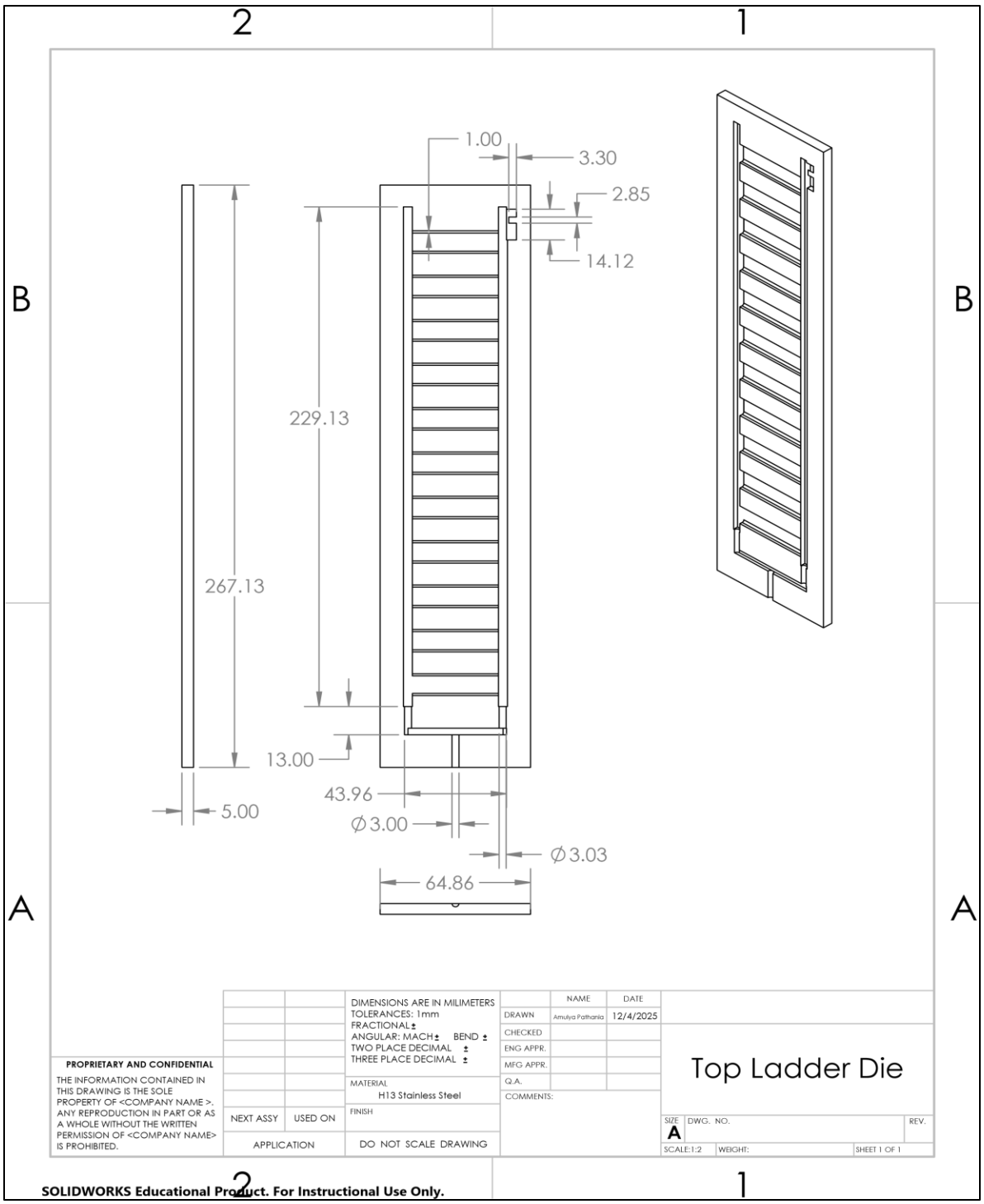


Figure 26: Engineering Drawing of Top Ladder Die



Figure 27: Sand-Cast Aluminum Wheels

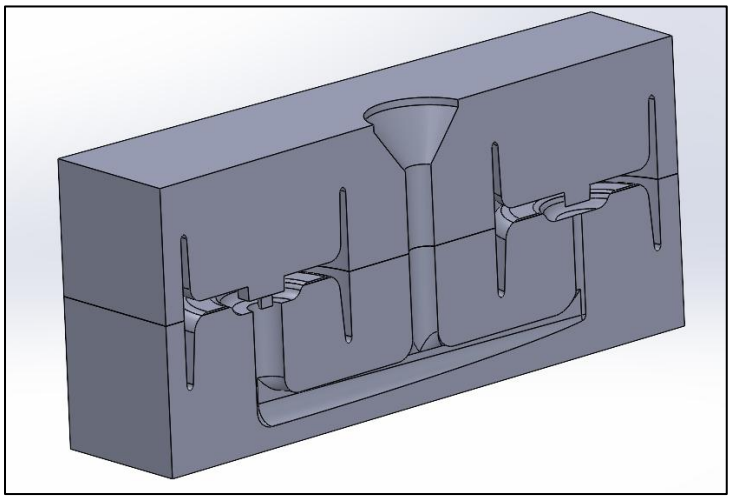


Figure 28: Section View of Wheel Die

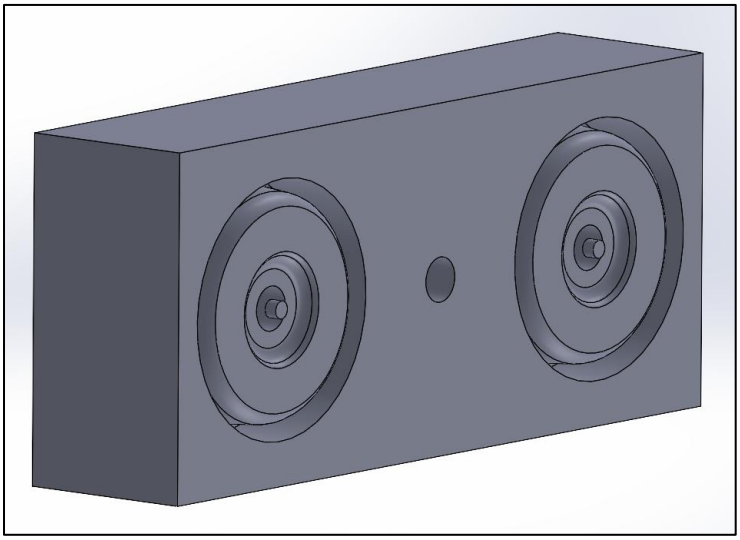


Figure 29: Bottom Wheel Die

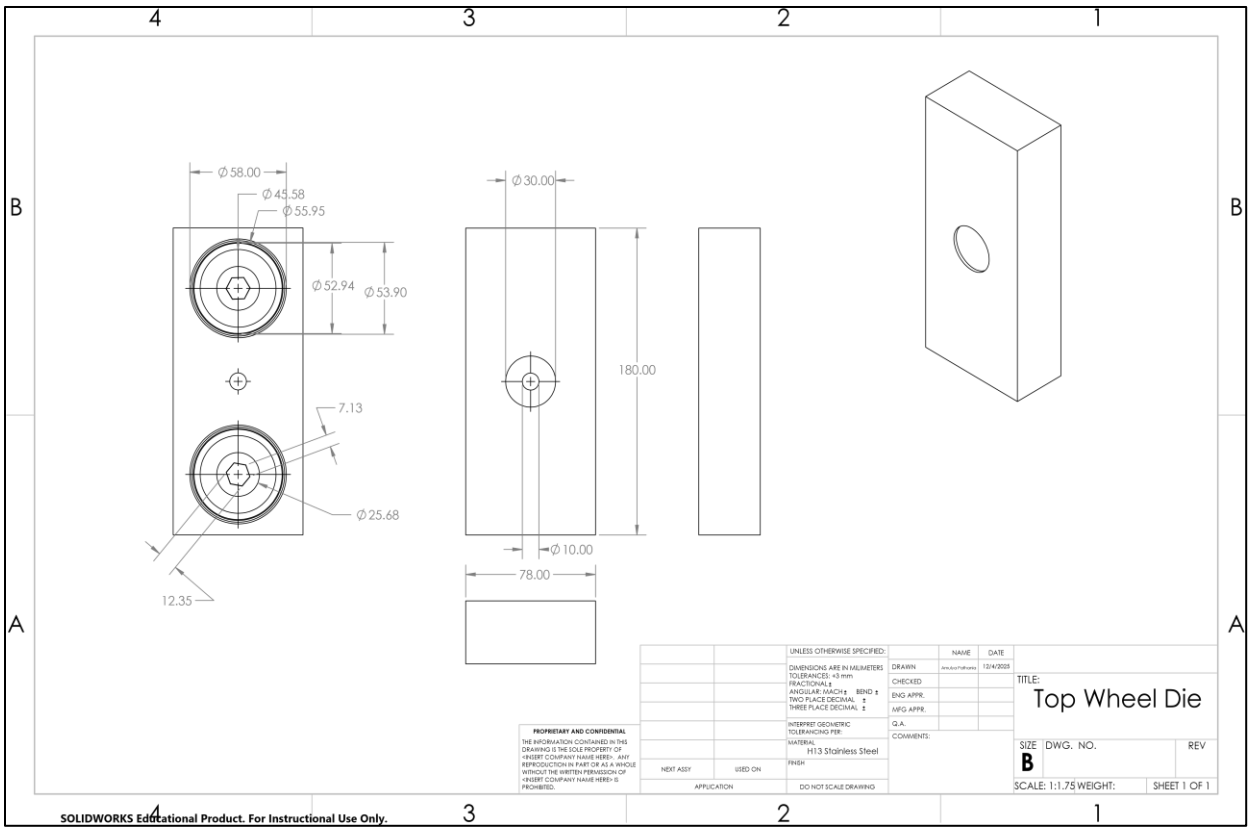


Figure 30: Engineering Drawing of Top Wheel Die

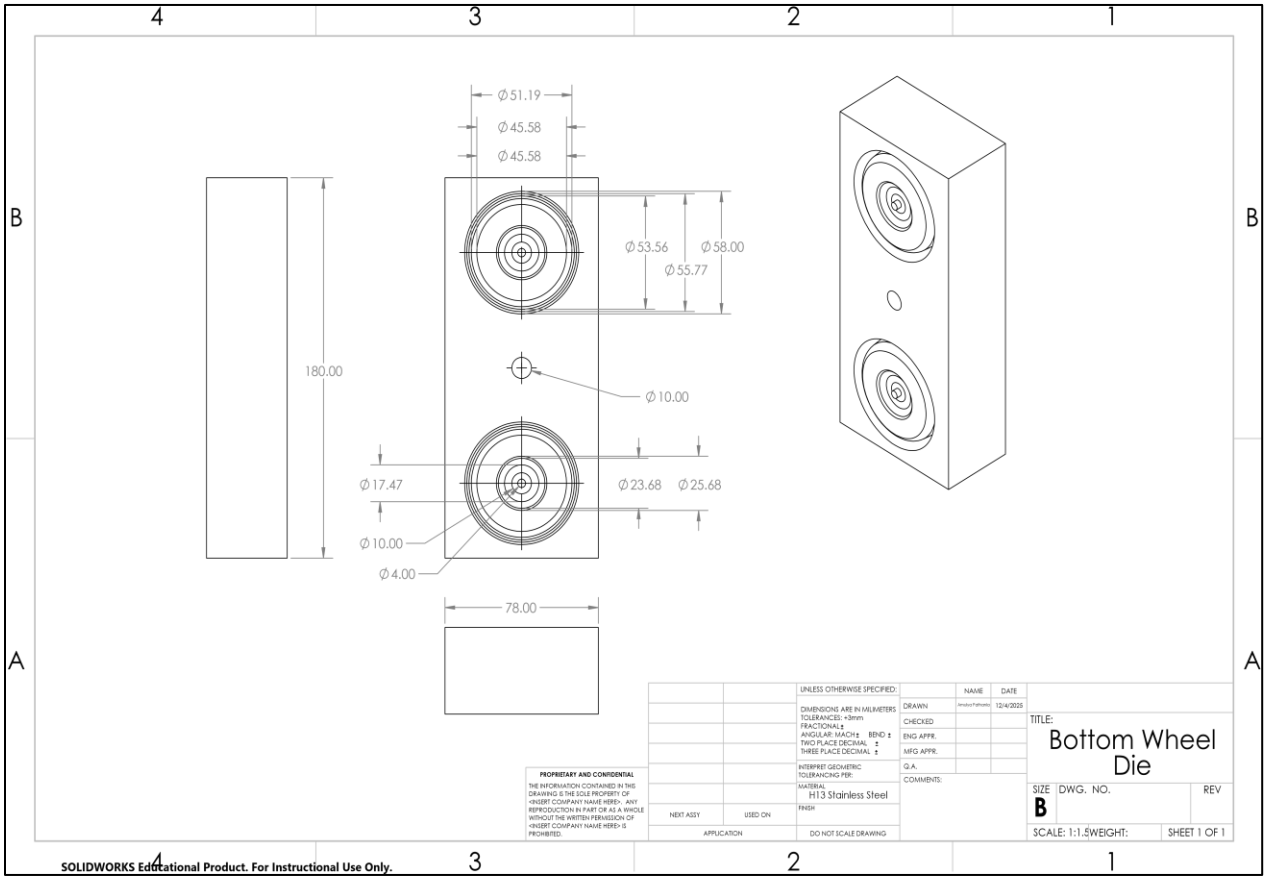


Figure 31: Engineering Drawing of Bottom Wheel Die